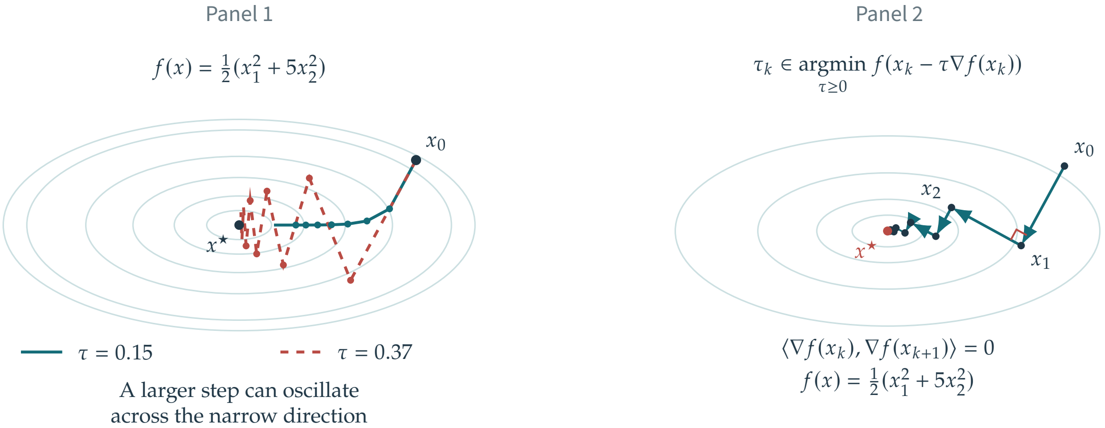

Panel 1
$f(x) = \frac{1}{2}(x_1^2 + 5x_2^2)$
x_0
$x^\star$
$\tau = 0.15$
$\tau = 0.37$
A larger step can oscillate across the narrow direction
Panel 2
$\tau_k \in \operatorname{argmin}_{\tau \geq 0} f(x_k - \tau \nabla f(x_k))$
x_0
x_1
x_2
$x^\star$
$\langle \nabla f(x_k), \nabla f(x_{k+1}) \rangle = 0$
$f(x) = \frac{1}{2}(x_1^2 + 5x_2^2)$